2026 edition

Deep Learning for Music Analysis and Generation

Music Transcription

(audio → scores)

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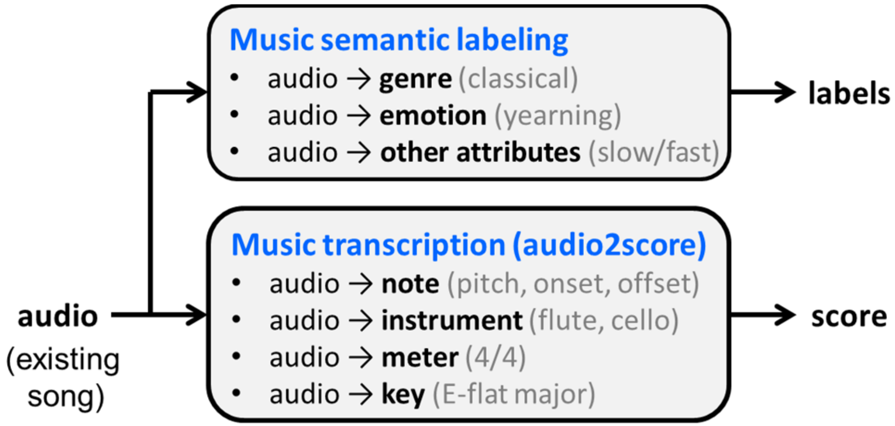
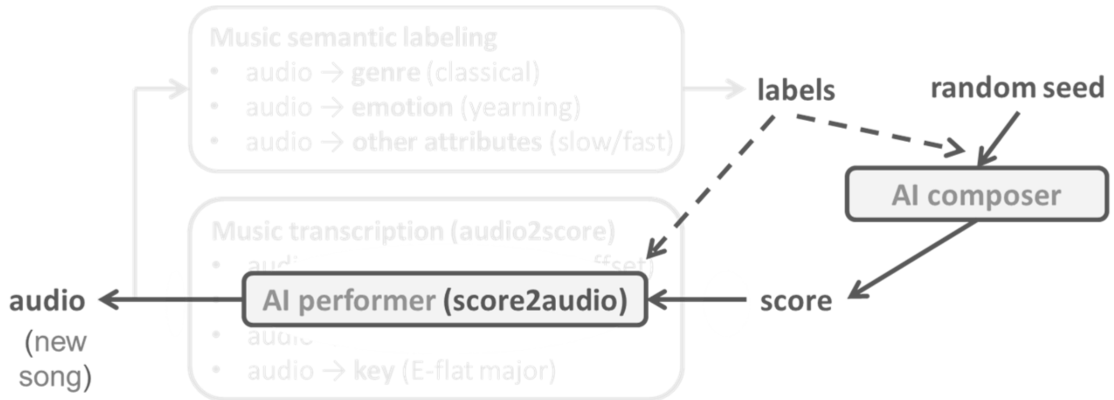
Music AI; or *Music Information Research* (MIR)

- **Music analysis**



  - music understanding
  - music search
  - music recommendation

- Music generation



  - MIDI generation
  - audio generation
  - MIDI-to-audio generation

Reference 1: FMP Notebook

https://www.audiolabs-erlangen.de/resources/MIR/FMP/C1/C1.html

Part	Title	Notions, Techniques & Algorithms	HTML	IPYNB
B	Basics	Basic information on Python, Jupyter notebooks, Anaconda package management system, Python environments, visualizations, and other topics	[html]	[ipynb]
0	Overview	Overview of the notebooks (https://www.audiolabs-erlangen.de/FMP)	[html]	[ipynb]
1	Music Representations	Music notation, MIDI, audio signal, waveform, pitch, loudness, timbre	[html]	[ipynb]
2	Fourier Analysis of Signals	Discrete/analog signal, sinusoid, exponential, Fourier transform, Fourier representation, DFT, FFT, STFT	[html]	[ipynb]
3	Music Synchronization	Chroma feature, dynamic programming, dynamic time warping (DTW), alignment, user interface	[html]	[ipynb]
4	Music Structure Analysis	Similarity matrix, repetition, thumbnail, homogeneity, novelty, evaluation, precision, recall, F-measure, visualization, scape plot	[html]	[ipynb]
5	Chord Recognition	Harmony, music theory, chords, scales, templates, hidden Markov model (HMM), evaluation	[html]	[ipynb]
6	Tempo and Beat Tracking	Onset, novelty, tempo, tempogram, beat, periodicity, Fourier analysis, autocorrelation	[html]	[ipynb]
7	Content-Based Audio Retrieval	Identification, fingerprint, indexing, inverted list, matching, version, cover song	[html]	[ipynb]
8	Musically Informed Audio Decomposition	Harmonic/percussive separation, signal reconstruction, instantaneous frequency, fundamental frequency (F0), trajectory, nonnegative matrix factorization (NMF)	[html]	[ipynb]

Reference 2: ISMIR 2018 & 2021 Tutorials

https://rachelbittner.weebly.com/other-resources.html

Tutorials

Programming MIR Baselines from Scratch: Three Case Studies 2021

International Society for Music InformationRetrieval (ISMIR) conference

- Part 1: Transcription with NMF (Ethan Manilow)
- Part 2: Pitch Tracking with pytorch (**Rachel Bittner**)
- Part 3: Instrument Classification with OpenL3 & Tensorflow (Mark Cartwright)

📺 See the recording here.

Fundamental Frequency Estimation in Music 2018

International Society for Music Information Retrieval (ISMIR) conference

- Part 1: Pitch (Alain de Cheveigné)
- Part 2: Polyphonic fundamental frequency estimation (**Rachel Bittner**)
- Part 3: Applications (Johana Devaney)

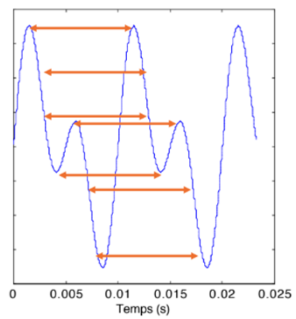
Monophonic F0 Estimation

- Part 1: Pitch (Alain de Cheveigné)

https://drive.google.com/file/d/1uWCwE03dM0j0Ptp1g5vPsT56nPd0DCW2/view

- Pitch: depends on fundamental frequency (F0), not on waveform shape
- **Frequency-domain F0 analysis**
  - For example: return the frequency bin that has the highest energy per frame
  - Problems: harmonics can be stronger, missing fundamental, etc
- Time-domain F0 analysis
  (based on **auto-correlation)**

```
librosa.pyin(y, *, fmin, fmax, sr=22050,
```



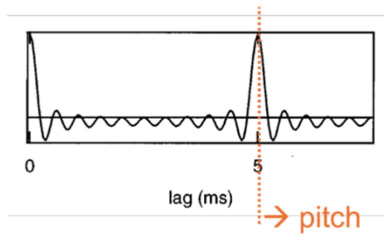
"shift and compare"

$$A_t(\tau) = \sum_W x(t)x(t-\tau)$$

product

time lag

integration window



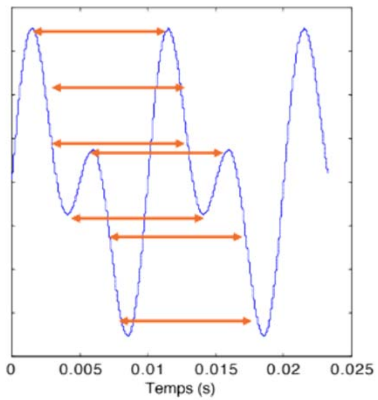
Monophonic F0 Estimation

• Part 1: Pitch (Alain de Cheveigné)

https://drive.google.com/file/d/1uWCwE03dM0j0Ptp1g5vPsT56nPd0DCW2/view

- Pitch: depends on fundamental frequency (F0), not on waveform shape
- Frequency-domain F0 analysis
- **Time-domain F0 analysis** (based on **auto-correlation)**
  - **YIN** (2002), **pYIN** (2014)

```
librosa.pyin(y, *, fmin, fmax, sr=22050,
```



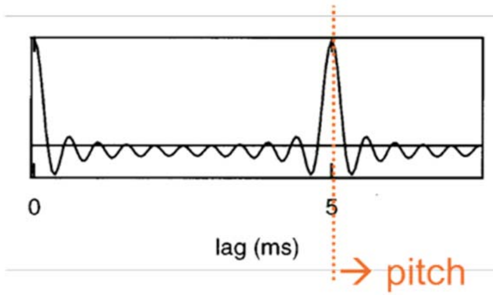
"shift and compare"

$$A_t(\tau) = \sum_{W} x(t)x(t-\tau)$$

product

time lag

integration window

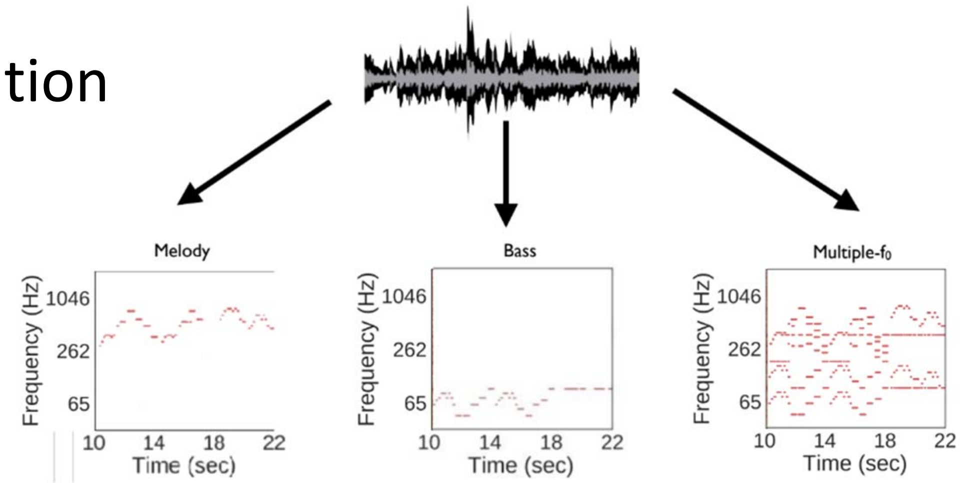


F0 Estimation

• Part 2: (**Rachel Bittner**)

https://drive.google.com/file/d/1Jmj6tNBGFMlElEQldgNZ2_YgD17gS6Sv/view

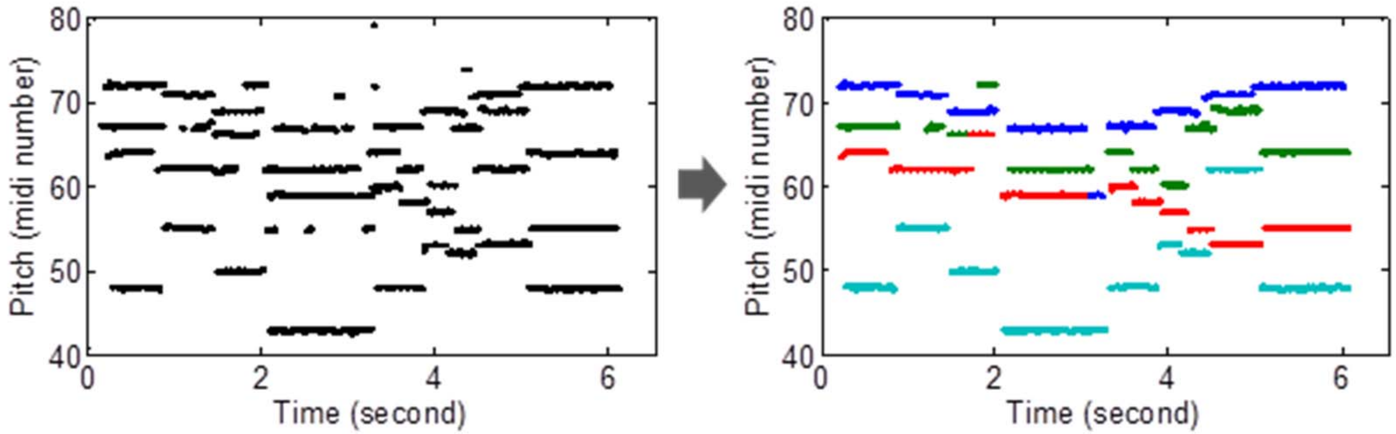
- **Monophonic** input vs **polyphonic** input
  - For example, singing solo vs. mixed song
- **Single F0 (melody)** vs **multiple F0** estimation
  - For example, vocal melody extraction
- **F0 estimation (hz)** vs
  note estimation (MIDI pitch)
- Time quantization or not
- Output instrument labels or not



F0 Estimation: Background

https://drive.google.com/file/d/1Jmj6tNBGFMlElEQldgNZ2_YgD17gS6Sv/view

- Related tasks
  - frame-level f0 estimation
  - note estimation
  - streaming
  - score transcription
- Time resolution
- Frequency resolution
- Voicing
- Instrument labeling



Melody Extraction vs. Note Transcription

- **Melody extraction**: F0 (can reflect *overshoot*, *vibrato*, *glissando*, etc)
- **Note transcription**: MIDI pitch (*quantized in frequency*)

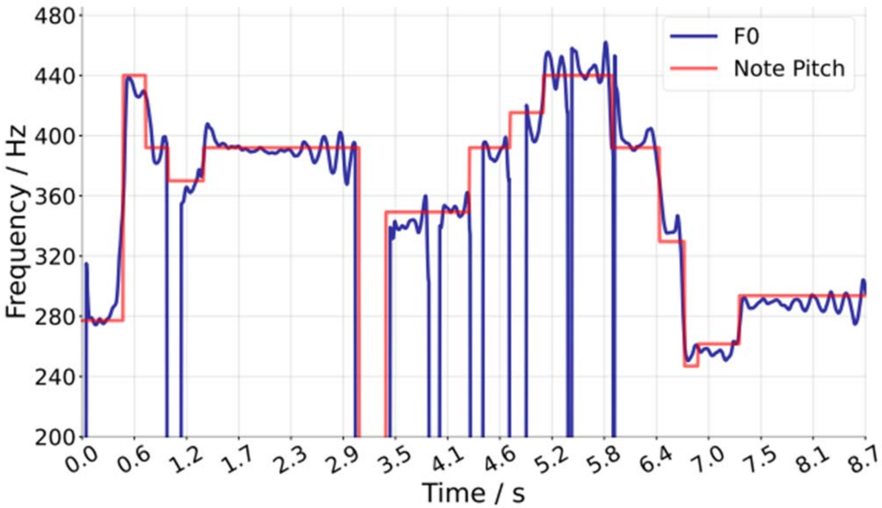


Figure source: M4Singer paper (https://openreview.net/pdf?id=qiDmAaG6mP)

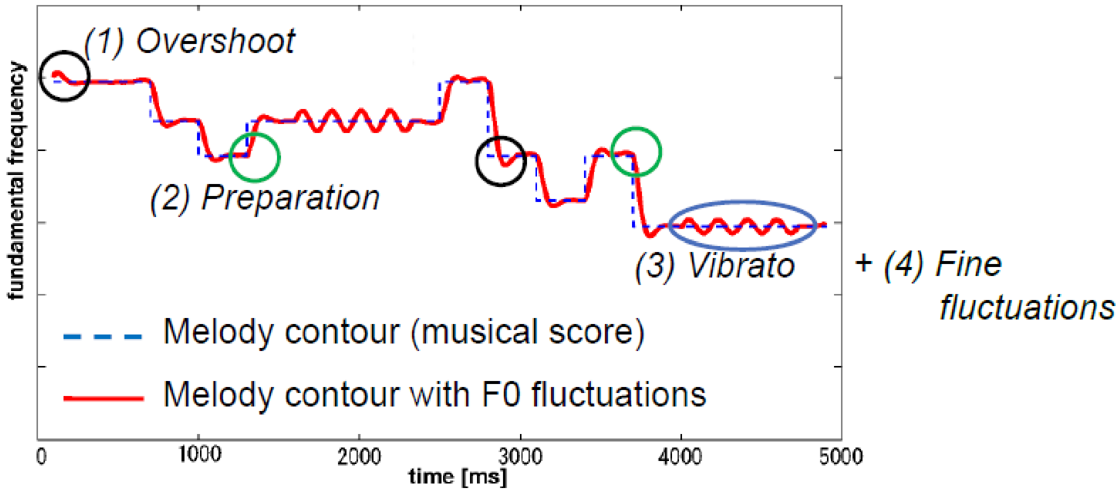


Figure from: Saitou et al, "Speech-to-singing synthesis: converting speaking voices to singing voices by controlling acoustic features unique to singing voices," WASPAA 2007

Pianoroll vs MusicXML

- **Time quantization**: quarter note, eighth note etc
- More than that…
- Usually, the target is to output a single-track or multi-track pianoroll

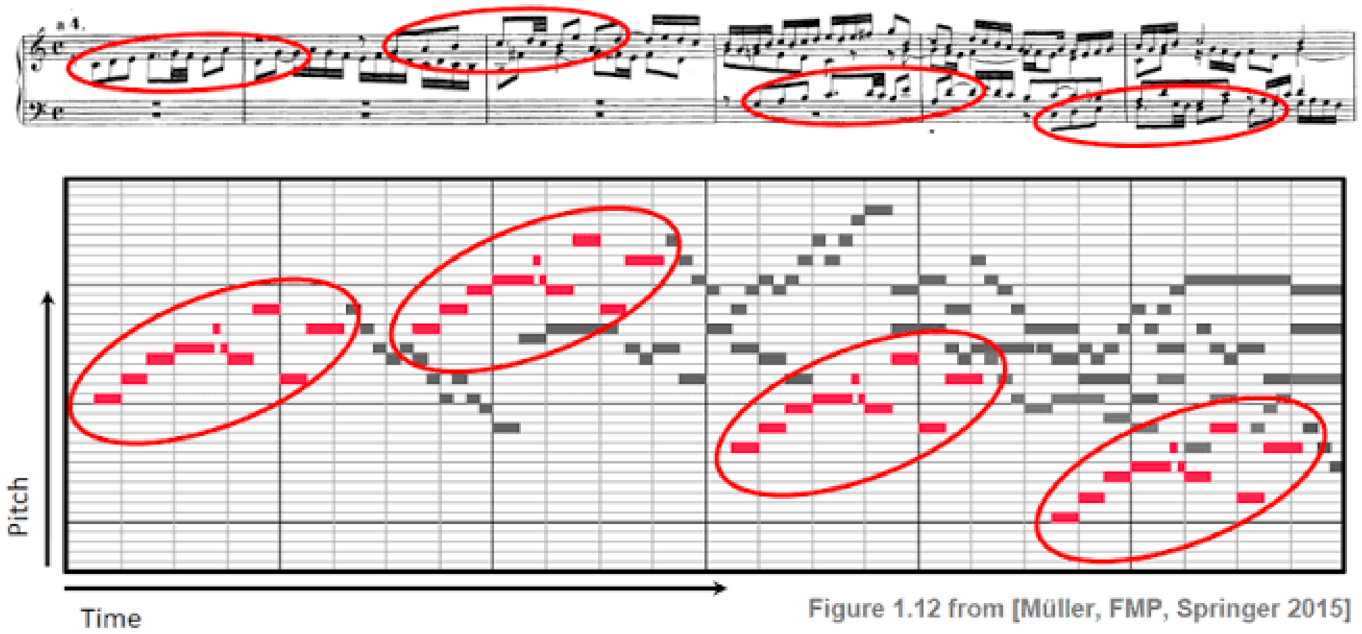
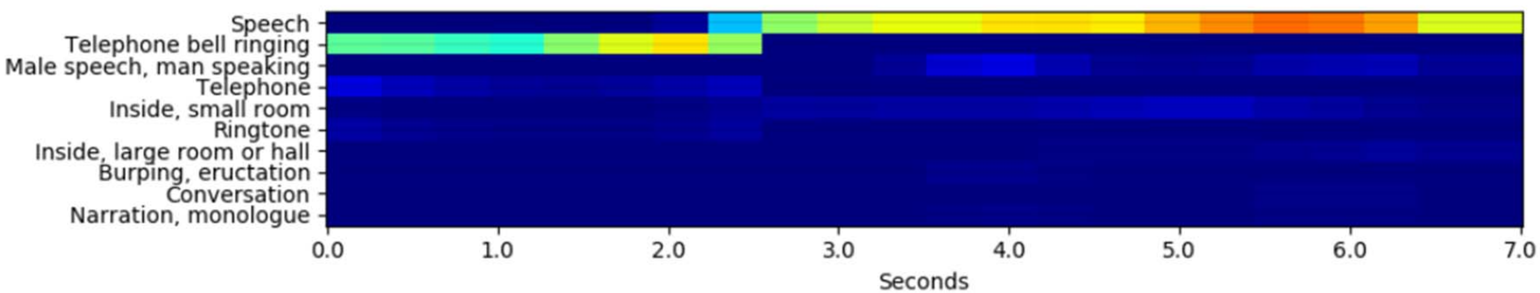
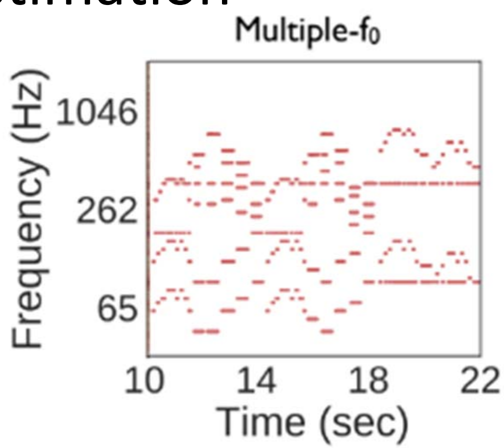


Figure 1.12 from [Müller, FMP, Springer 2015]

From an ML/DL Viewpoint

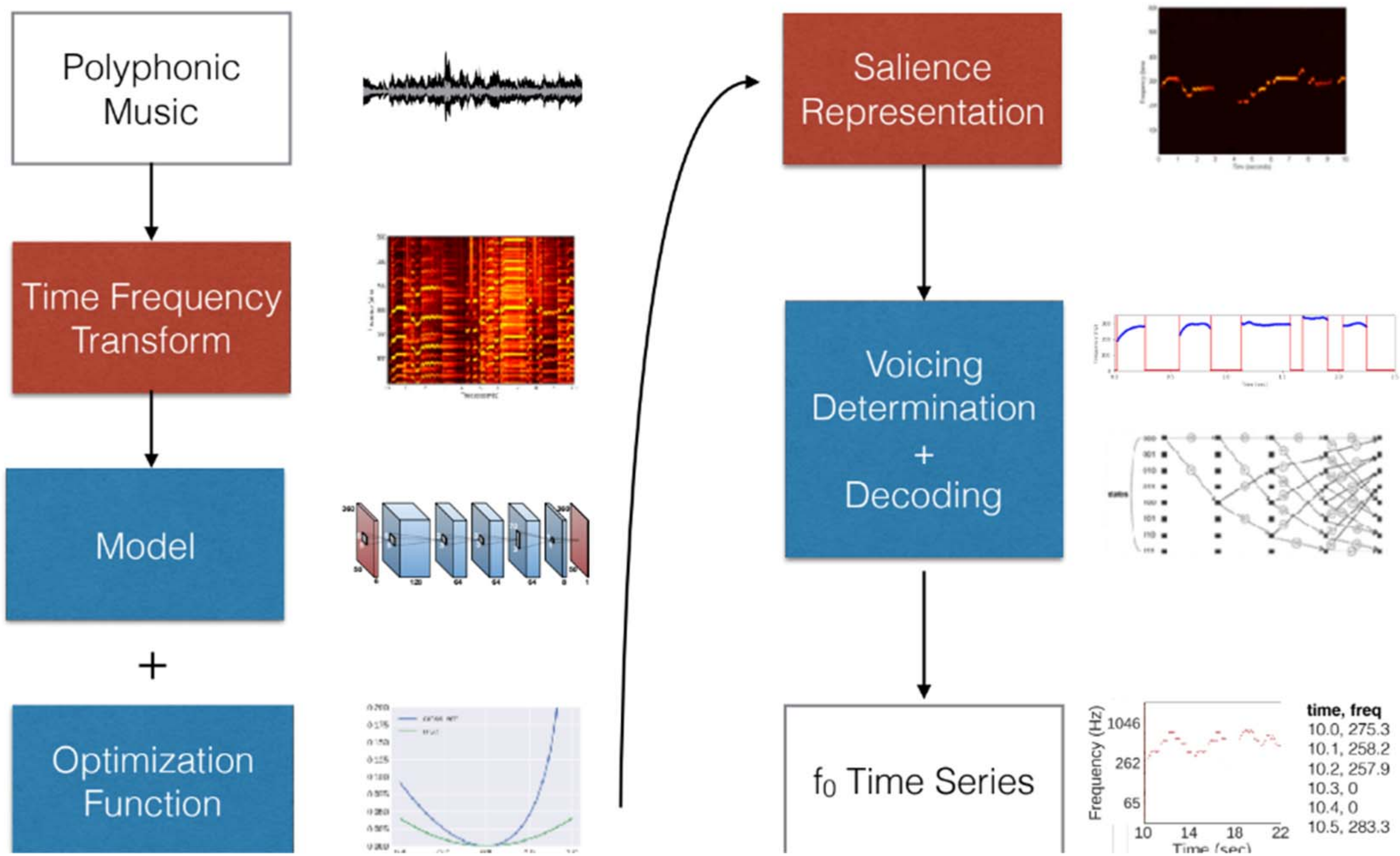
- **Per song:** genre classification
  - we can make predictions for each chunk and then aggregate the result over time (e.g., by taking the average logits or by majority voting on the chunk-level decisions)
- **Per short-time chunk:** audio event detection, instrument activity detection
  - e.g., output is a matrix [class x time]
- **Per time-frequency point:** f0 estimation, multi-pitch estimation
  - e.g., output is a matrix [frequency x time]





Polyphonic F0 Estimation: Typical Approach

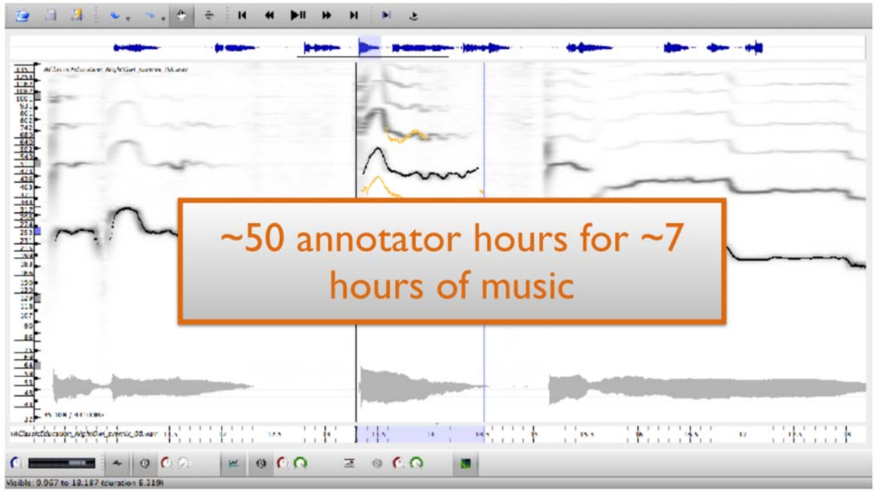
https://drive.google.com/file/d/1Jmj6tNBGFMlElEQldgNZ2_YgD17gS6Sv/view

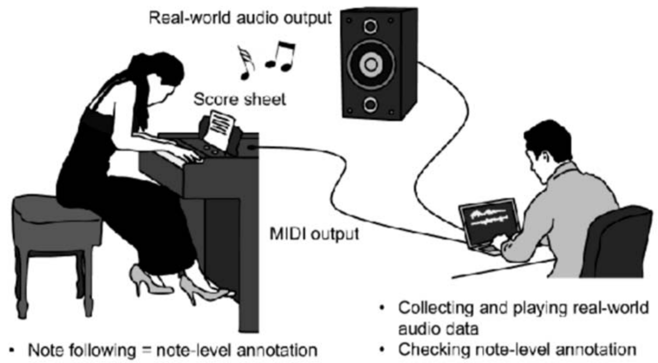


Supervised Training

https://drive.google.com/file/d/1Jmj6tNBGFMlElEQldgNZ2_YgD17gS6Sv/view

- But labeled data is hard to collect
- Slow down research progress
  - Single F0 estimation for mono signals is nearly solved
  - Multi-F0 estimation still has room for improvement
  - Note transcription for arbitrary instruments remains hard
    - One exception is piano note transcription, which is nearly solved, partly thanks to high-quality piano synthesizers





Guitar Transcription: Need to be Invariant to Audio Effects

(Examples provided by Positive Grid®)

Dry (single coil EG)

In the Clouds

Overdriven Verb Icon

Lazy Down

Chen et al., "Towards automatic transcription of polyphonic electric guitar music: A new dataset and a multi-loss transformer model," ICASSP 2022

Evaluation Metrics for Pitch-Related Tasks

https://craffel.github.io/mir_eval/

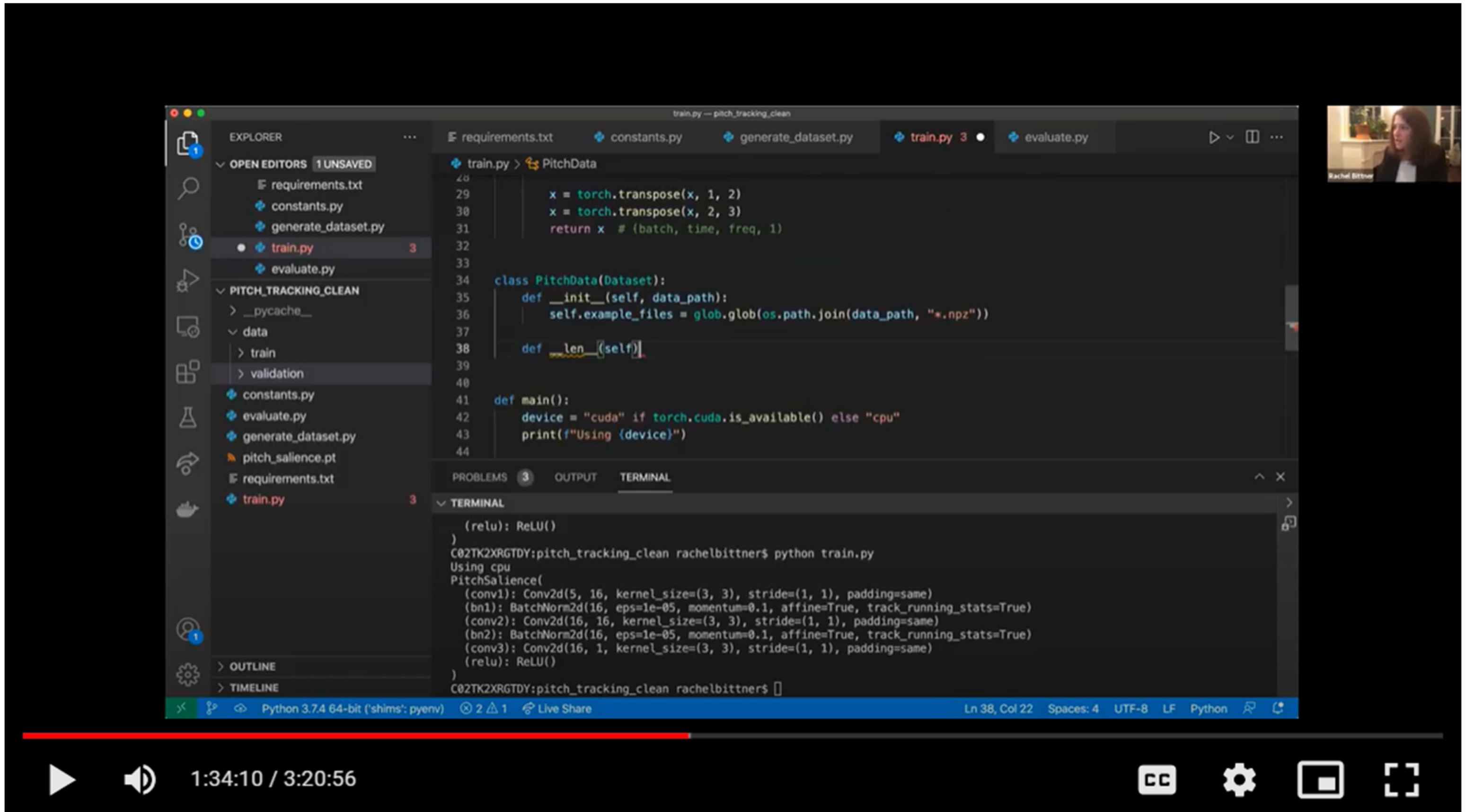
- **Single- and multi-pitch estimation**
  - correct if within 0.5 semitones of a reference frequency
  - chroma accuracy (ignore octave)
  - voicing measures
- **Piano transcription**
  - onset tolerance window: 50ms
  - offset tolerance window: 20% of note duration
  - onset only vs. onset+offset
- **Chord transcription**
  - root
  - majmin
  - majmin_inv
  - thirds
  - triads
  - tetrads
  - sevenths
  - overseg, underseg, seg

ISMIR 2021 Tutorial: Programming MIR Baselines from Scratch - Pitch Tracking

https://github.com/rabitt/ismir-2021-tutorial-case-studies/tree/main/pitch_tracking

- Video online

  https://drive.google.com/file/d/18LNaKy2ymFjEWj19gHy25wgtFV4pdML0/view

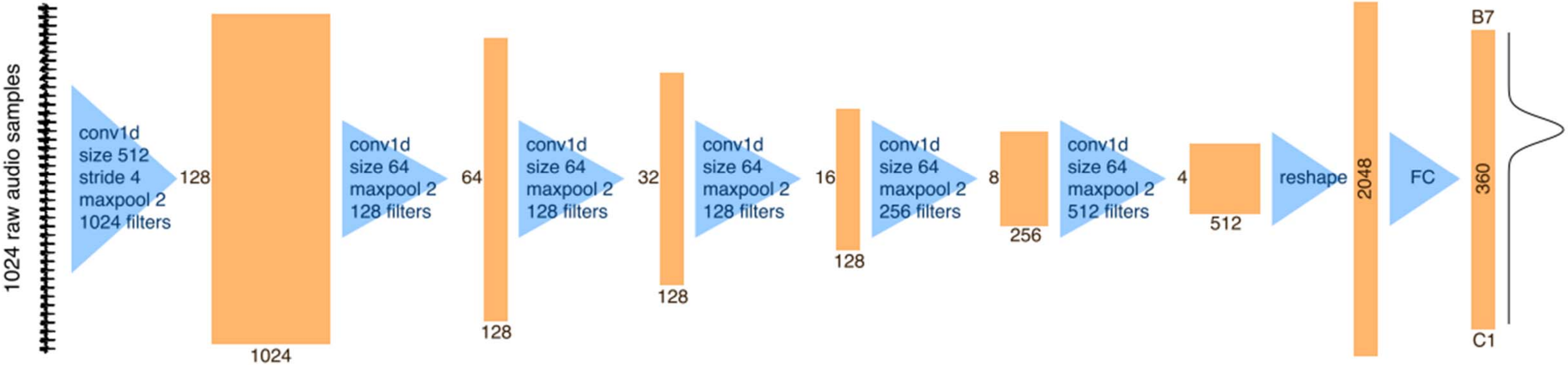


Tasks & Exemplar Models

- **Melody extraction**
  - F0 estimation in monophonic music
  - F0 estimation in polyphonic music
- **Leadsheet recognition**: quantized melody + chord
- **Multipitch estimation & multi-instrument transcription**

Melody Extraction from "Monophonic" Music: CREPE

https://github.com/marl/crepe



- Make a prediction every 1,024 samples
- Freq resolution: 20 cents (1/5 semitone)
- Outperform DSP-based methods such as pYIN and SWIPE

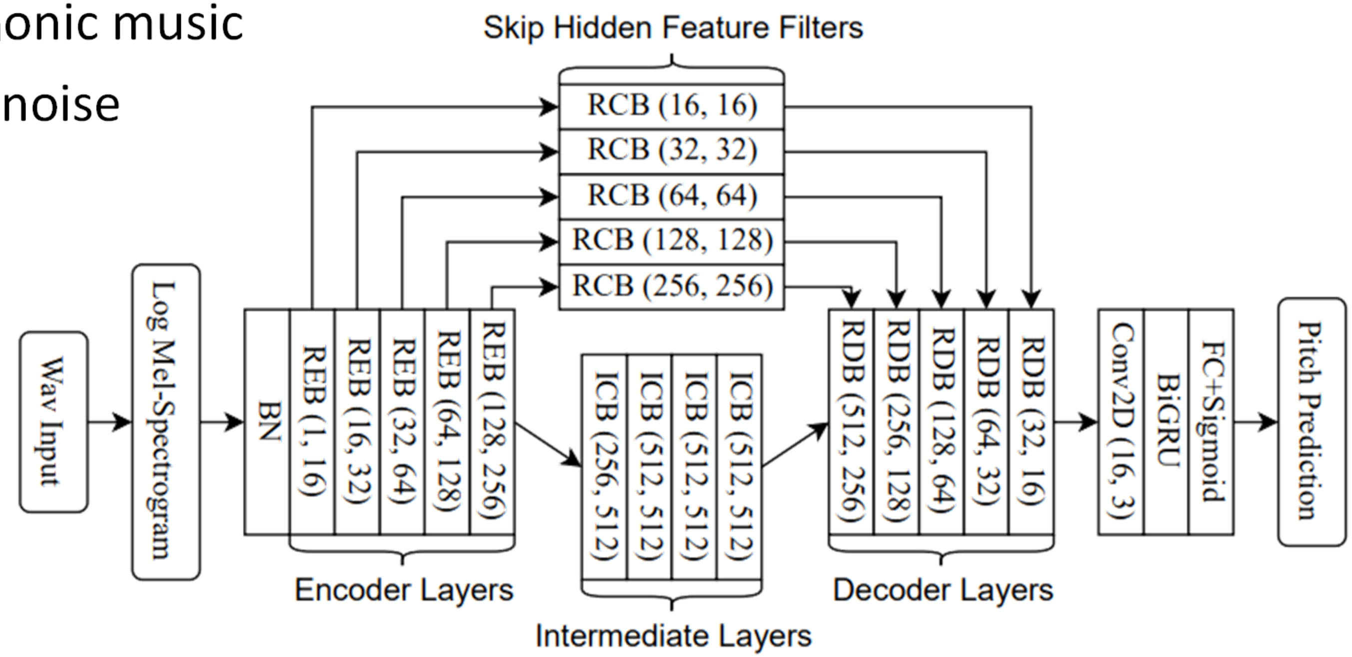
Dataset	Threshold	CREPE	pYIN	SWIPE
RWC-synth	50 cents	**0.999±0.002**	0.990±0.006	0.963±0.023
	25 cents	**0.999±0.003**	0.972±0.012	0.949±0.026
	10 cents	**0.995±0.004**	0.908±0.032	0.833±0.055

Kim et al., "CREPE: A convolutional representation for pitch estimation," ICASSP 2018

Vocal Melody Extraction from “Polyphonic” Music: RMVPE

https://github.com/Dream-High/RMVPE

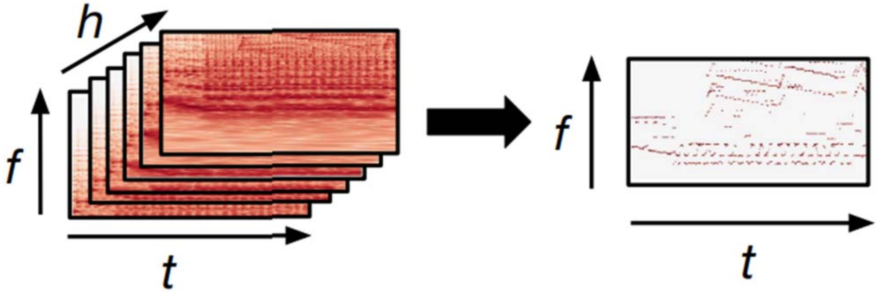
- Use deep U-Net and GRU to directly extract **vocal** f0 from polyphonic music
- Robust to different types of noise

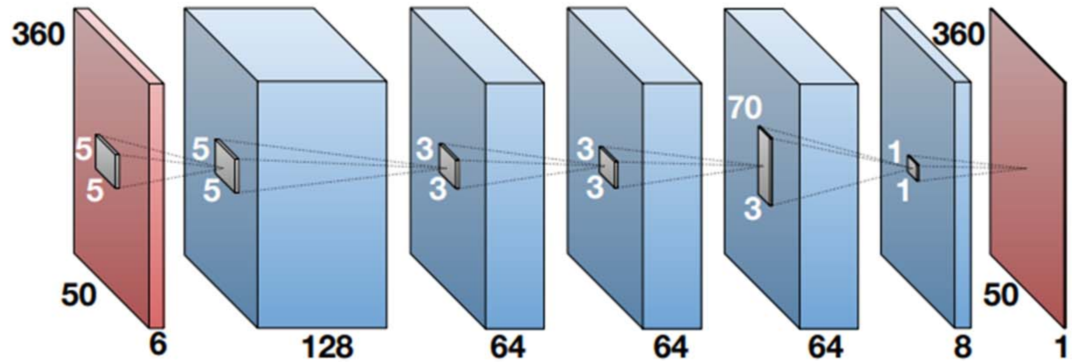


Wei et al., “RMVPE: A robust model for vocal pitch estimation in polyphonic music,” INTERSPEECH 2023

Melody Extraction and Multi-F0 Estimation: DeepSalience

https://github.com/rabitt/ismir2017-deepsalience





- Input: harmonic constant-Q transform (HCQT) with 6 channels
- Model: 5-layer CNN (**no strides, no pooling → no shift-invariance**)
  - (5 x 5): 1 semitone in frequency and 50 ms in time
  - (70 x 3): 14 semitones in frequency to capture relationships between frequency content within an octave

Kim et al., "Deep salience representations for F0 estimation in polyphonic music," ISMIR 2017

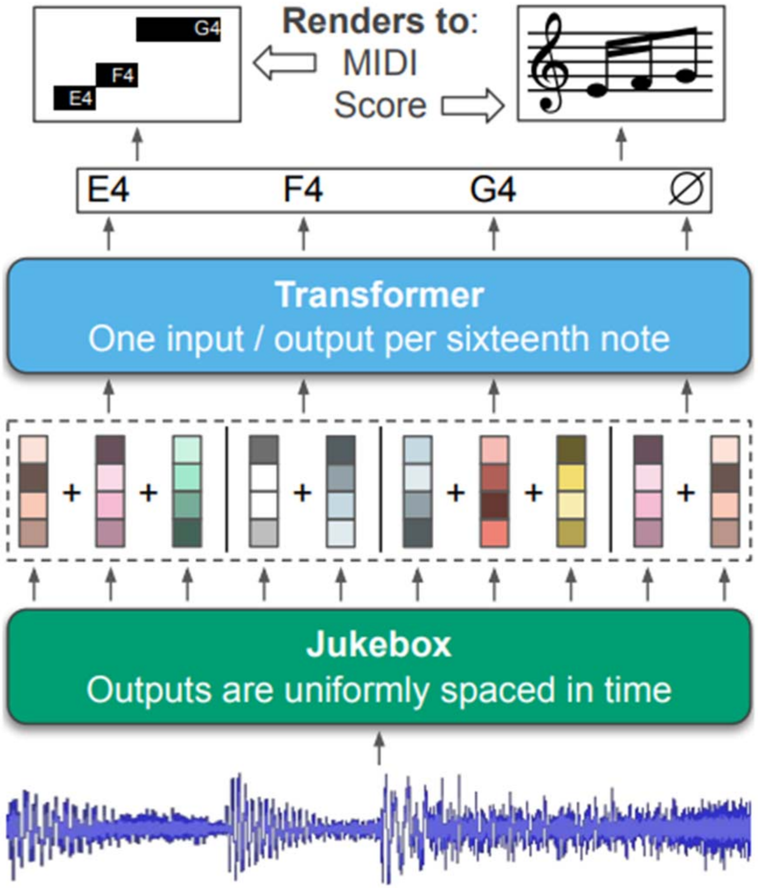
Tasks & Exemplar Models

- **Melody extraction**
- **Leadsheet recognition**: “quantized” melody + chord
- **Multipitch estimation & multi-instrument transcription**

Lead Sheet Transcription: Sheet Sage

https://github.com/chrisdonahue/sheetsage

- Predict both melody and chord
  - Use a large pre-trained model called **Jukebox** as model backbone and then do transfer learning
  - Use **Transformer** to learn the language model (LM) for melody and chords
  - Computationally heavy but pretty accurate
  - Lighter alternatives: BTC (https://github.com/jayg996/BTC-ISMIR19)



Donahue et al., "Melody transcription via generative pre-training," ISMIR 2022

Lead Sheet Transcription: Sheet Sage-Pro

https://github.com/pingw220/sheetsage-pro

- Component-wise upgrade
  - "Mel-Band RoFormer" for vocal separation
  - "Beat This!" for beat/downbeat tracking
  - "GAME" for melody extraction
  - Jiang's model for chord recognition
  - "All-in-One" for structure/meter analysis
- Output melody+chord+**lyrics**
  - Lyrics **with timestamps**
  - Via lyrics transcription or audio-lyrics alignment

Task	Metric	Sheet Sage	SheetSage-Pro
Melody, ZH	Note F1@50 ↑	0.049	**0.250**
Melody, EN	Note F1@50 ↑	0.112	0.340
ST500, ZH	Note F1@50 ↑	0.400	0.578
ST500, ZH	Onset F1@50 ↑	0.514	0.645
Chords	MajMin WCSR ↑	0.621	**0.904**
Chord bounds	Boundary F1 ↑	0.559	**0.799**
Lyrics, ZH	Syll. onset ↓	—	**24 ms**
Lyrics, EN	Word onset ↓	—	**33 ms**

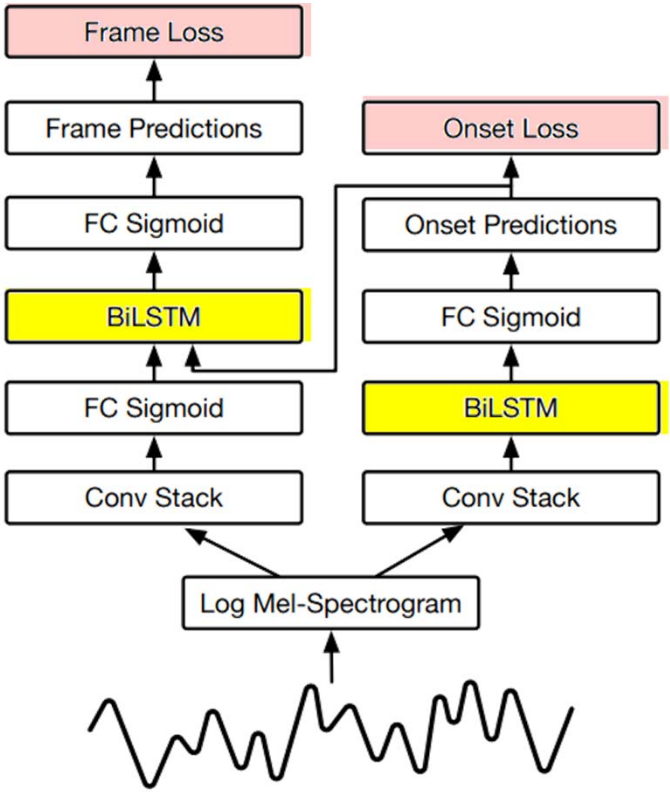
Wang et al., "SheetSage-Pro: Bilingual lyric-aligned lead-sheet transcription from music audio," ISMIR-LBD 2026

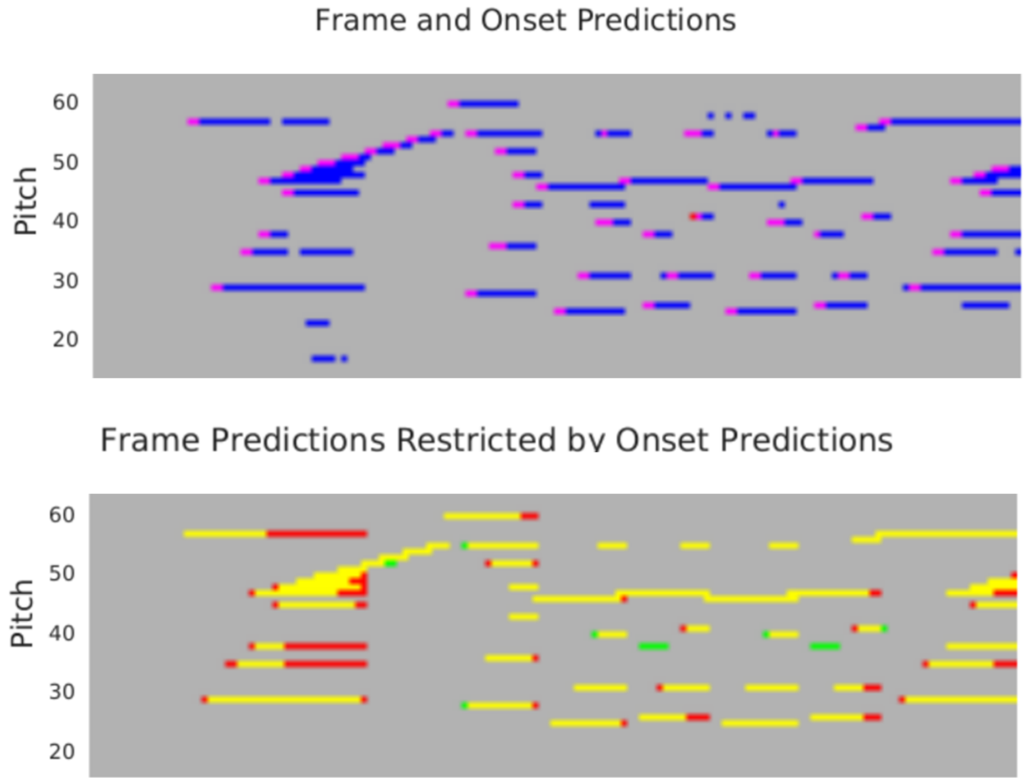
Tasks & Exemplar Models

- **Melody extraction**
- **Leadsheet recognition**: quantized melody + chord
- **Multipitch estimation & multi-instrument transcription**
  - Multipitch estimation in single instrument (piano; output is a single-track pianoroll)
  - Multipitch estimation in general music (output is a single-track pianoroll)
  - Multi-instrument transcription (output is a multi-track pianoroll)

Piano Transcription: Onset-and-Frames

https://magenta.tensorflow.org/onsets-frames



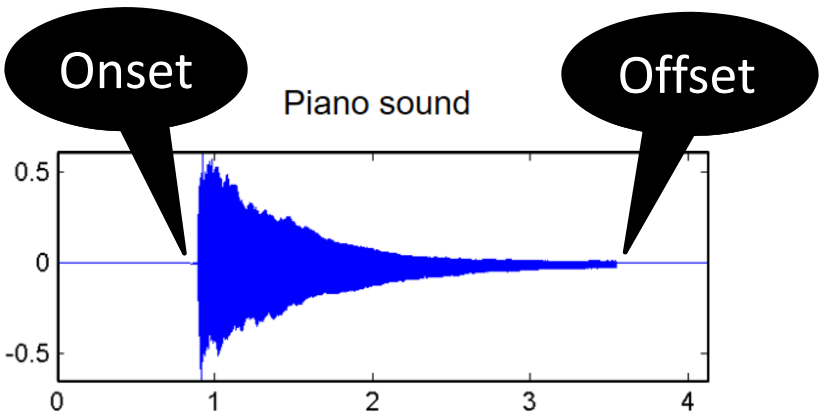


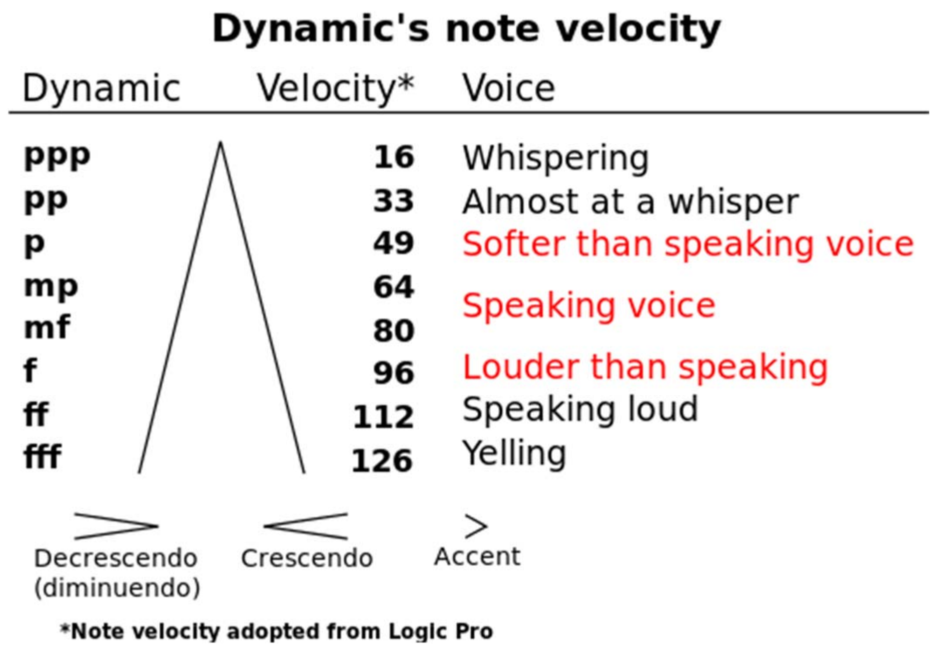
Hawthorne et al., “Onsets and Frames: Dual-objective piano transcription,” ISMIR 2018

Pitch, Onset, Offset, Velocity

https://magenta.tensorflow.org/datasets/maestro

"We partnered with organizers of the International **Piano-e-Competition** for the raw data used in this dataset. During each installment of the competition virtuoso pianists perform on **Yamaha Disklaviers** which, in addition to being concert-quality acoustic grand pianos, utilize an integrated high-precision **MIDI capture** and playback system."

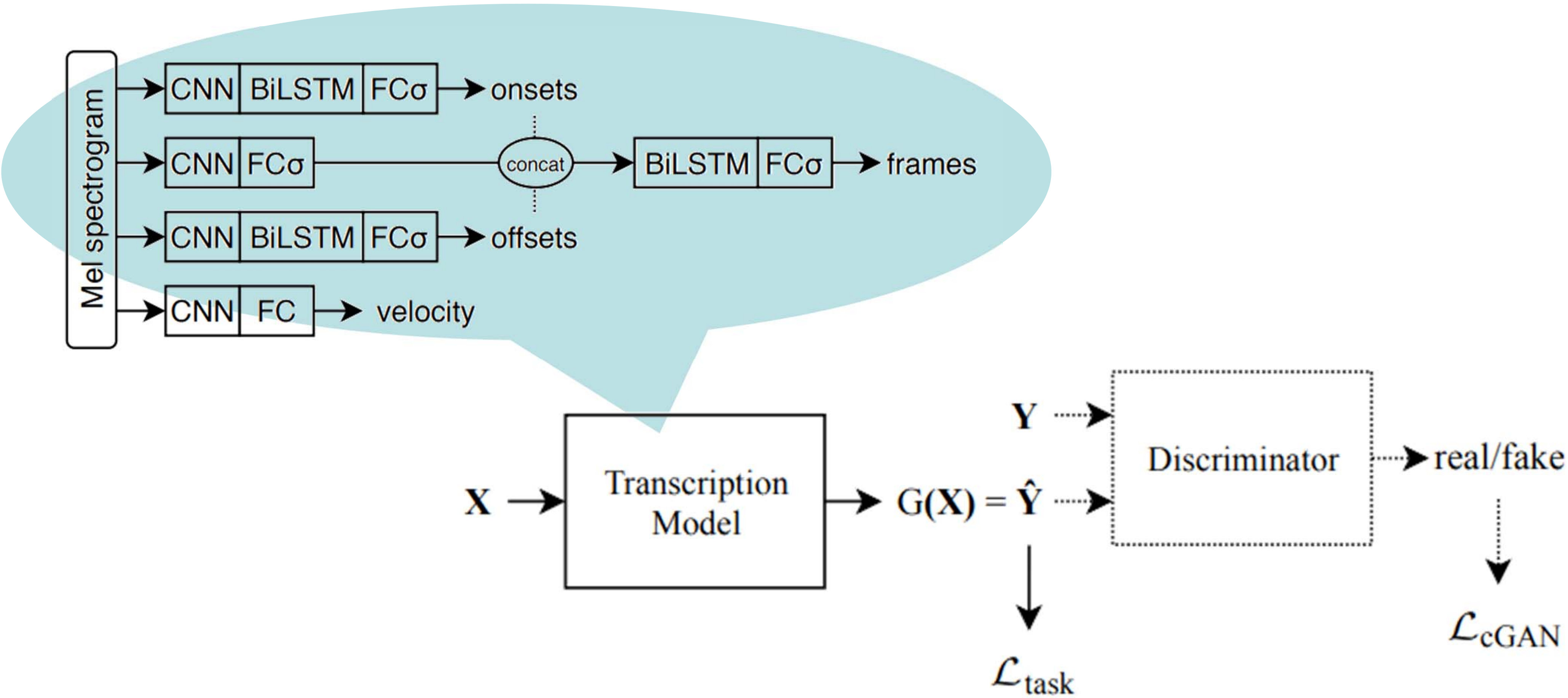




https://freeonlinesheetmusic.wordpress.com/2016/01/10/dynamics/

Hawthorne et al., "Enabling factorized piano music modeling and generation with the MAESTRO dataset," ICLR 2019

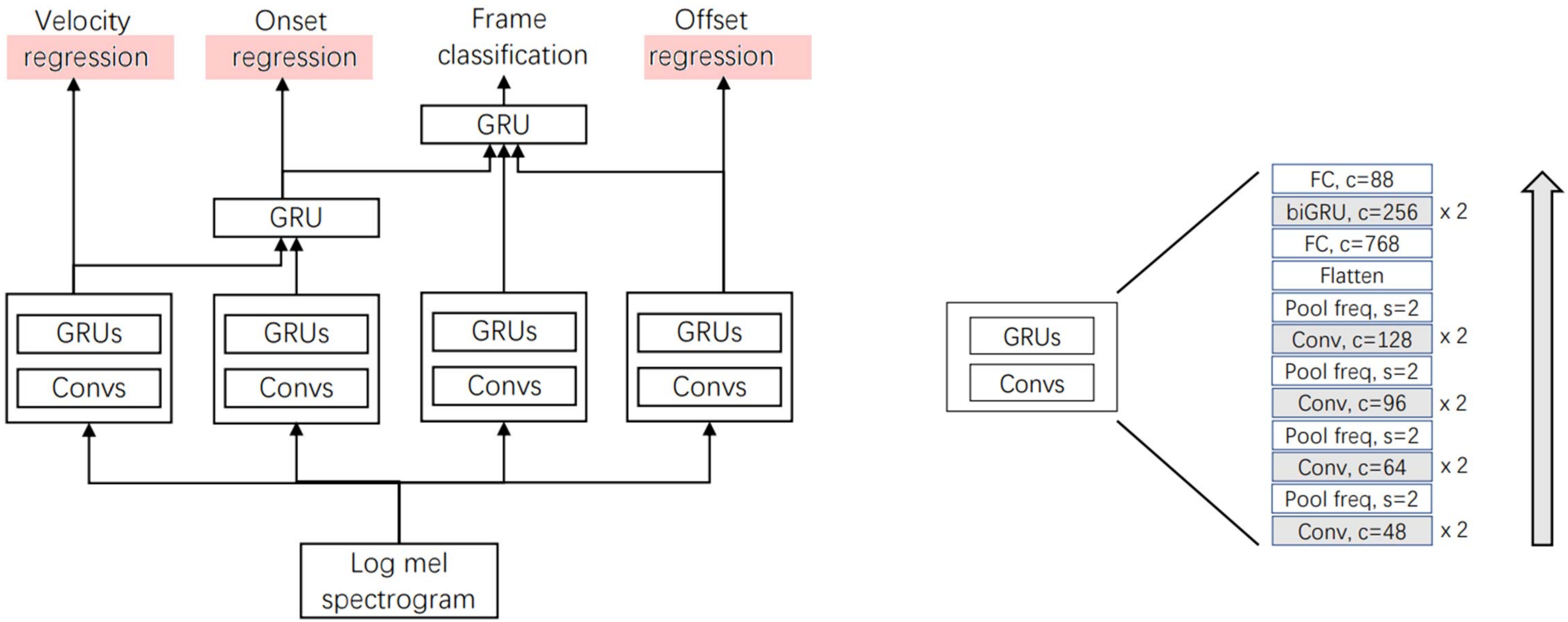
Piano Transcription with GAN



Kim & Bello, "Adversarial learning for improved onsets and frames music transcription," ISMIR 2019

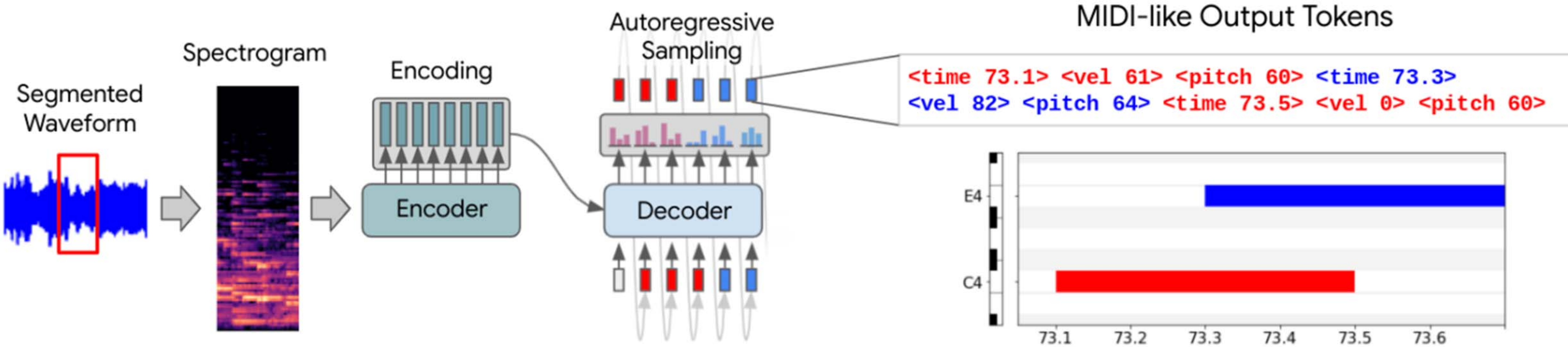
High-resolution Piano Transcription

https://github.com/bytedance/piano_transcription



Kong et al., “High-resolution piano transcription with pedals by regressing onsets and offsets times,” TASLP 2021

Seq2Seq Transformers for Piano Transcription



- Jointly modeling audio features and language-like output dependencies
- Possible to pre-train the decoder

	Model	Onset, Offset, & Velocity F1	Onset & Offset F1	Onset F1
MAESTRO V1.0.0	Transformer (ours)	**82.18**	**83.46**	95.95
	Kong et al. 2020 [19]	80.92	82.47	**96.72**
	Kwon et al. 2020 [21]	–	79.36	94.67
	Kim & Bello 2019 [20]	80.20	81.30	95.60
	Hawthorne et al. 2019 [2]	77.54	80.50	95.32

Hawthorne et al., "Sequence-to-sequence piano transcription with Transformers," ISMIR 2021

Note Transcription from Polyphonic Music: Basic Pitch

https://github.com/spotify/basic-pitch

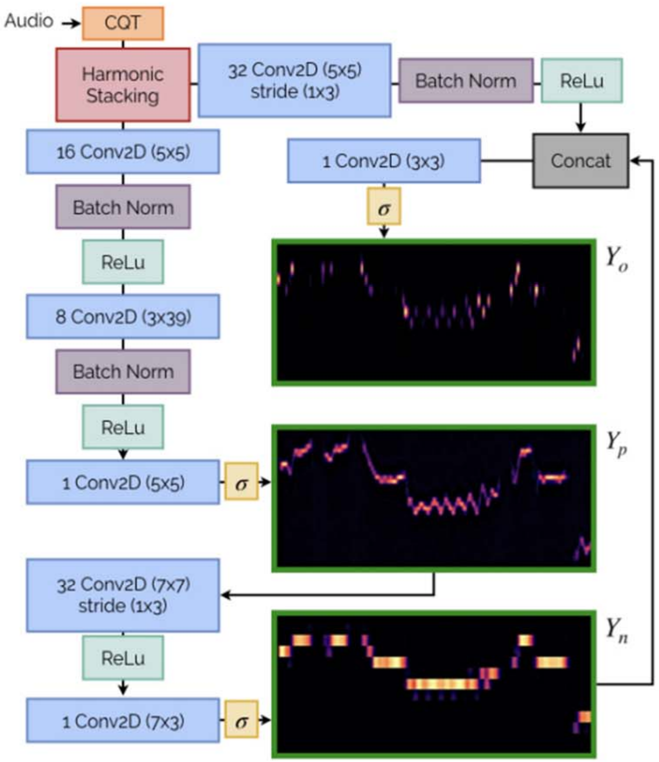


Fig. 1. The NMP architecture. The matrix posteriorgram outputs Y_o, Y_p, and Y_n are outlined in green. σ indicates a sigmoid activation.

- For *instrument-agnostic* note transcription and multi-pitch estimation
- Pure CNN-based for being light-weight

Try Basic Pitch, a free audio-to-MIDI converter with pitch bend detection, built by Spotify. Learn more or follow the instructions below.

1 — Press record and sing a ditty into your computer. Or drop a recording of any single instrument (piano, guitar, xylophone, you name it).

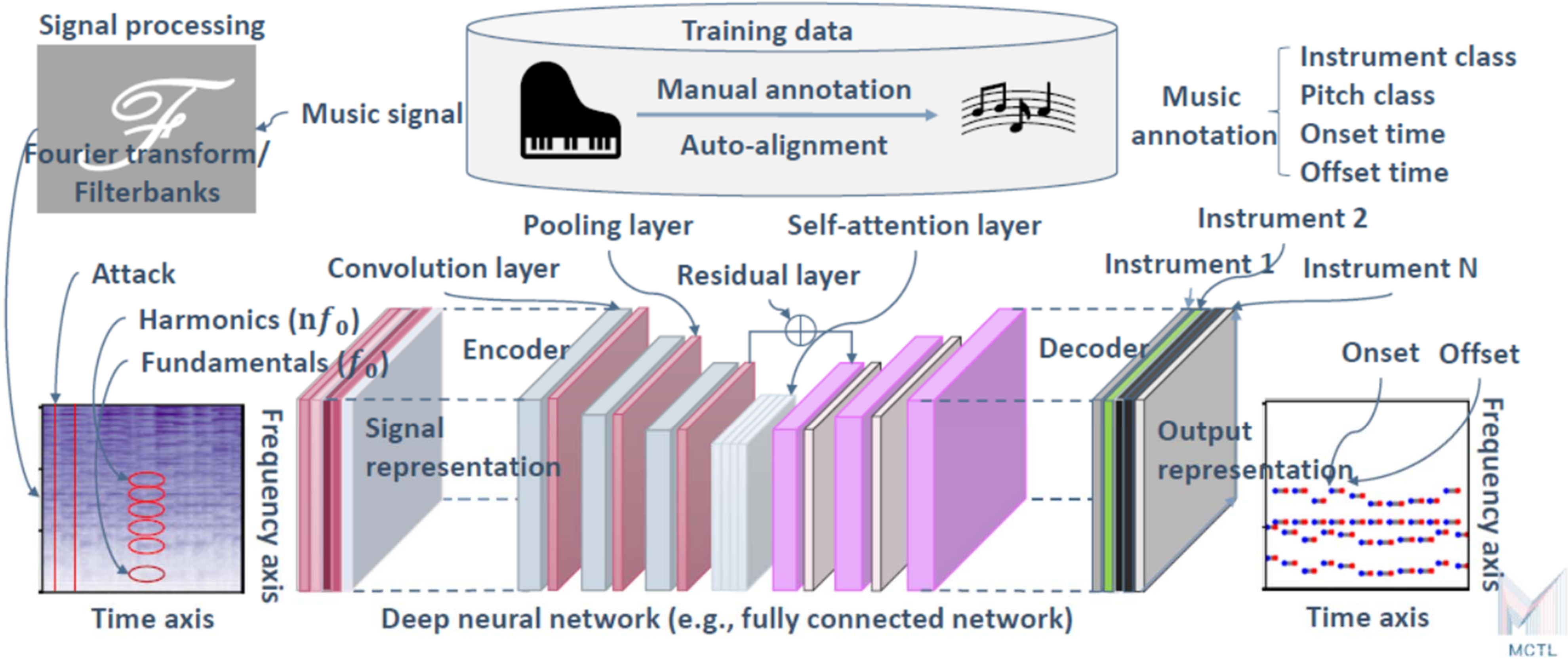
2 — Then get a MIDI version back. Just like that.

3 — Download the MIDI file to fine tune and make corrections in your favorite digital audio workstation.

Bittner et al., "A lightweight instrument-agnostic model for polyphonic note transcription and multipitch estimation," ICASSP 2022

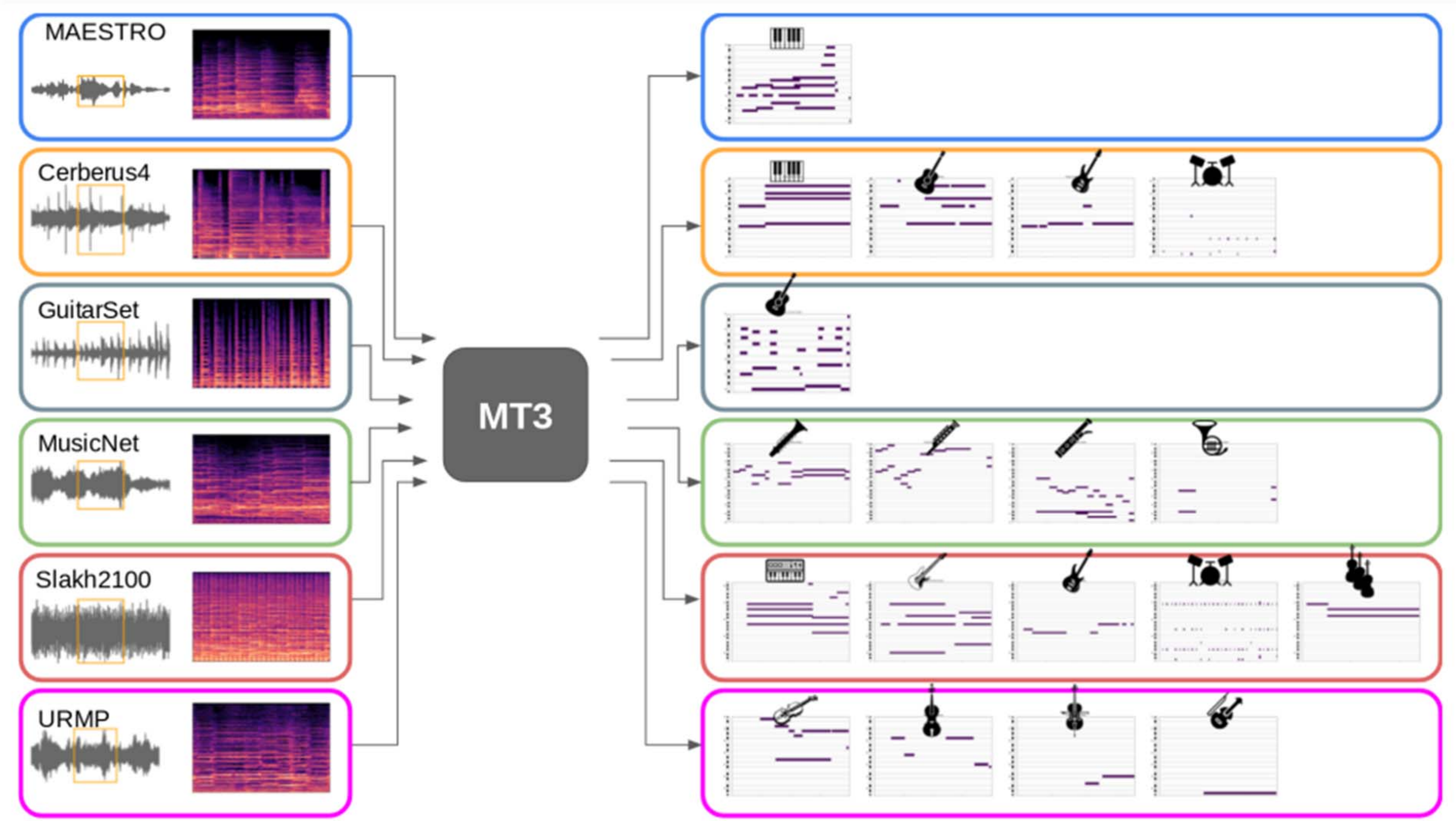
Multi-Instrument Transcription: Omnizart

https://github.com/Music-and-Culture-Technology-Lab/omnizart



Wu et al., "Omnizart: A general toolbox for automatic music transcription," JOOS 2021

Multi-Instrument Transcription: MT3

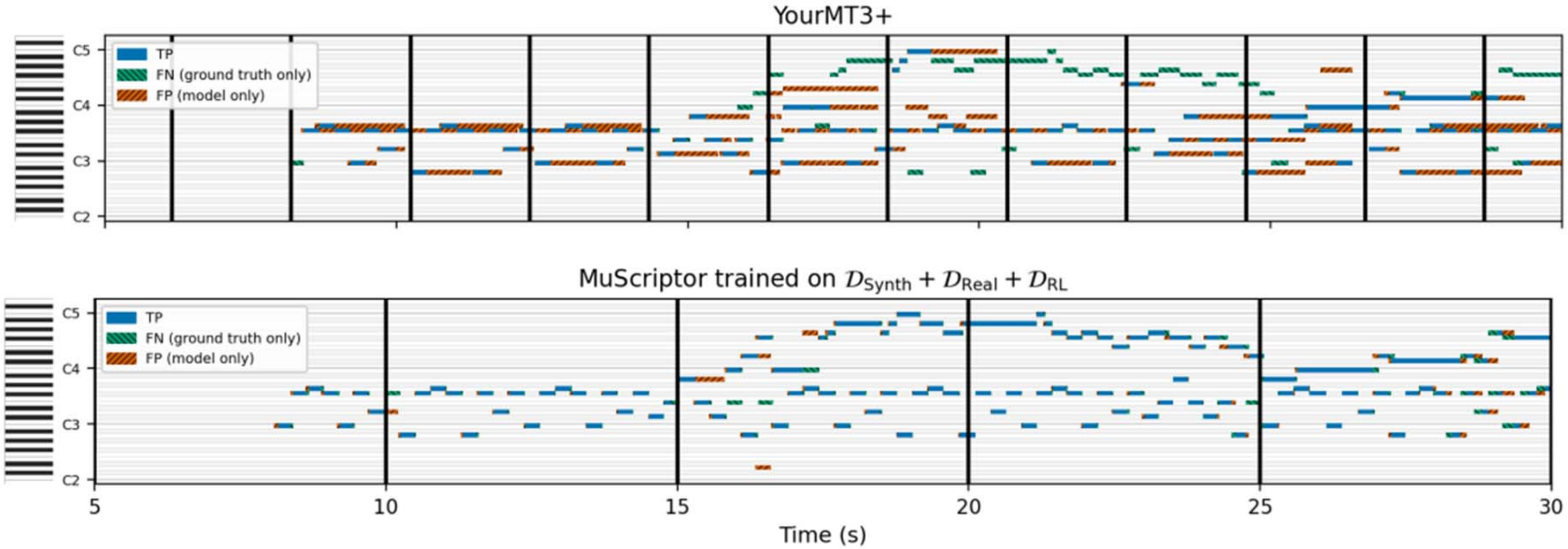


Gardner et al., “MT3: Multi-task multitrack music transcription,” ICLR 2022

Multi-Instrument Transcription: MuseScriptor

https://github.com/muscriptor/muscriptor

- SOTA model trained with 170k recordings (11k hours) with aligned note annotations, alongside a synthetic dataset of 1.45M MIDIs
- Excellent result when instrument tags are given



Rouard et al., “MuScriptor: An Open model for multi-instrument music transcription,” ISMIR 2026